EVEREST ENGINEERING COLLEGE

(AFFILIATED TO POKHARA UNIVERSITY

A Minor Project Proposal

On

STOCK MARKET ANALYSIS AND FORECASTING

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ABSTRACT

Stock market forecasting is one of the biggest challenges in the financial market since its time series has a complex, noisy, chaotic, dynamic, volatile, and non-parametric nature. However, due to computing development, an intelligent model can help investors and professional analysts reduce the risk of their investments. As *Machine Learning* models have been extensively studied in recent years, several studies have explored these techniques to predict stock prices using historical data and technical indicators. However, as the objective is to generate forecasts for the financial market, it naturally becomes important that the forecast should be profitable. Therefore, this systematic review focuses on Machine Learning models implemented for stock market forecasting using technical analysis. Discussions were made based on four main points of view: predictor techniques, trading strategies, profitability metrics, and risk management. This study showed that the *LSTM* technique is widely applied in this scenario (73.5%). This work significant contribution is to highlight some limitations found in the literature, such as only 35.3% of the studies analysed profitability, and only two articles implemented risk management. Therefore, despite the widely explored theme, there are still interesting open areas for research and development.

Table of Contents

ACKNOWLEDGEMENT i

ABSTRACT ii

Table of Contents iii

List of Figures v

List of Tables vi

ABBREVIATIONS vii

Chapter 1: INTRODUCTION 1

1.1. Background and Problem Statement 1

1.2. Objectives 2

1.3. Scope and Applications 2

Chapter 2: LITERATURE REVIEW 4

2.1. Deep Learning for Finance: Deep Portfolios 4

2.2. ARIMA Model for Forecasting Stock Prices 4

2.3. The Performance of LSTM and BiLSTM in Forecasting Time Series 5

2.4. NSE Stock Market Prediction Using Deep-Learning Models 5

2.5. Stock Price Prediction Using Machine Learning and LSTM-Based Deep Learning Models 5

Chapter 3: METHODOLOGY 6

3.1. System Block Diagram 6

3.1.1. Data Collection and Preprocessing 6

3.1.2. LSTM Model Architecture 7

3.1.3. Model Training and Evaluation 7

3.2. Description of working principle 8

3.3. System Development Life Cycle (SDLC) 9

3.4. Assumptions 10

3.5. Hardware and Software Required 10

3.6. Machine Learning Algorithms 11

3.7. Data Collection 11

3.8. Implementation Plan 11

Chapter 4: Expected Output 12

4.1.1 Performance Metrics 12

Chapter 5: Project Schedule 13

5.1. Schedule 13

Chapter 6: Feasibility Analysis 14

6.1. Technical Feasibility 14

6.2. Operational Feasibility 14

6.3. Economic Feasibility 14

6.4. Time Feasibility 15

Chapter 7: CONCLUSION 16

REFERENCES 17

List of Figures

Fig. No.	Title	Page Number
Figure 3.1	Block Diagram of Stock Prediction using LSTM model	6
Figure 4.1	Actual Price vs Predicted Price	11
Figure 3.3.1	Iterative and Incremental Model	9

List of Tables

Table Number	Title	Page Number
Table 5.1	Gantt Chart	13

ABBREVIATIONS

LSTM	Long Short-Term Memory
ANN	Artificial Neural Network
DL	Deep Learning
FA	Fundamental Analysis
TA	Technical Analysis
ARIMA	Autoregressive Integrated Moving Average
MA	Moving Average
ML	Machine Learning

Chapter 1: INTRODUCTION

Asset prices forecasting for stock market is a very difficult and complicated task since several micro and macroeconomic attributes and events influence the price formation, such as political events, news, company balance sheets, among others. These factors contribute to the nonlinear and non-stationary characteristics presented by the market, favouring the proposed task complexity.

Therefore, the studies of these influences are made through market analysis, and their main objective is to predict future directions to assist decision making based on market behaviour. The literature presents two main approaches: Fundamental Analysis (FA) and Technical Analysis (TA). Both have the same primary objective, and the difference is the information set used for forecasting and decision making. The first focuses on studying company data and seeks to determine whether it has growth potential in the medium to long term.

In contrast, TA doesn't consider the company data that much, because they want to buy at cheap and sell at high price for short period of time. In other words, company balance sheets, accounting scandals, financial crises, or any relevant information capable of generating volatility in an asset is reflected in their price. Therefore, it is possible to avoid the FA data, which are often subjective, to identify patterns present in the asset graph through this strategy type. Based on this principle, we are going to predict the future market based on the technical indicator such as closing price, MA etc.

1.1. Background and Problem Statement

Technical analysts make extensive use of Technical Indicators (TI) and candlestick pattern analysis to assist in price movement forecast. Several scientific articles used price information (Open, High, Low, Close prices – OHLC), trading volume, and indicators set as model input based on these techniques. However, when modelling these analyses, two different approaches are used; the works that used TIs generally adopted regression

techniques and those that analysed candlestick patterns adopted image processing techniques.

Even though looking through all the past data manually by ourself and identify the pattern is bit of a hassle. Instead, we can simply train the model to identify and recognize the market trend and fluctuations based on past data and predict the future prices based on these past data.

1.2. Objectives

The objectives of this study are twofold: first, to conduct a comprehensive analysis of the historical stock prices of these technology giants, and second, to develop and evaluate an LSTM-based prediction model that can forecast future stock prices with a high degree of accuracy. By achieving these objectives, this research seeks to contribute to the growing body of knowledge in the field of financial analysis and provide valuable insights for investors and policymakers and to describe the stock market and its components in day-to-day trading

- Develop a Predictive Model: To design and implement a Long Short-Term Memory (LSTM) neural network capable of accurately predicting stock prices based on historical data.
- Analyse Stock Market Trends: To analyse past stock market trends and patterns to identify significant factors influencing stock price movements.
- Improve Forecasting Accuracy: To enhance the forecasting accuracy of stock prices by leveraging the ability of LSTM to capture long-term dependencies in time series data.

1.3. Scope and Applications

- **Stock Market Forecasting:** The LSTM model can be applied to predict short-term and long-term stock price movements, aiding

traders, investors, and financial institutions in making informed decisions.

- **Risk Management**: By predicting potential price fluctuations, the model can help risk managers identify periods of high market volatility and adjust strategies accordingly.
- **Market Movement Analysis**: Combined with additional data sources, such as news or social media sentiment, the LSTM model can be used to analyze how external factors impact stock prices.
- **Financial Advisory Services**: Financial advisors and consultants can incorporate the model's forecasts into their advisory services, offering clients data-driven insights into market trends.

Chapter 2: LITERATURE REVIEW

This review takes lot of study and analysis into its domain field which is deep learning, the strategy of trading and the stock market data.

2.1. Deep Learning for Finance: Deep Portfolios

J.B. Heaton et.al [4] explore deep learning algorithm to solve problem that are arising in financial prediction. It also stated that how deep learning predictor is more efficient and productive in getting result than traditional predictors. It also clarifies that it is easy to handle correlation and how to avoid over fitting.

In this research world it is easily be seen that there are lot of deep learning algorithm. So finding the perfect deep learning algorithm for prediction is a mammoth task. It requires lot of study to find which is the best. Our study also requires to study about different type of algorithm and to see which is the best model. Torkil Aamodt find that Convolutional neural network had efficient result.

2.2. ARIMA Model for Forecasting Stock Prices

Mondale et.al[5] study and apply the Arima model to predict the accuracy of stock price prediction. This model was identified by using Akaike information criteria and it is found if there is change in training dataset than the variation in accuracy is little. To determine the accuracy, mean absolute error is the efficient way. LSTM had been proven successfully for time series prediction. Hengijian jia et al. proposed that the LSTM algorithm learn the stock price pattern in effective way and by applying this approach it gets lower MAE and RMSE value. This study helps to identify this problem as time series prediction and to use sliding window technique to get the better result.

2.3. The Performance of LSTM and BiLSTM in Forecasting Time Series

Sima Siami-Namini [6] in his research found that LSTM overcomes the Arima based model by a large margin and it is approx. 84 and 87 percent. The research also clarifies that when we expand the number of iteration then it does not increase the model efficiency in the stock price prediction. It has been reported that artificial Recurrent Neural Networks (RNN) with memory, such as Long Short-Term Memory (LSTM), are superior compared to Autoregressive Integrated Moving Average (ARIMA) with a large margin.

2.4. NSE Stock Market Prediction Using Deep-Learning Models

Hiransha[7] used an approach for predicting stock data using RNN, CNN deep learning architecture. This model is capable to detect the trends in stock market. This model gives the better result than the Arima model and in this study, it also clarifies that deep learning model is a best and efficient way to predict the time series data. Literature survey presents that the domain of stock price prediction is yet to be explored at depth as there are many more state-of-the-art techniques that has been proved a better option to predict the stock price.

2.5. Stock Price Prediction Using Machine Learning and LSTM-Based Deep Learning Models

Mehtab [8] used an approach where they took the NIFTY 50 index values of National Stock Exchange of India, during the period of December 29, 2014 till July 31, 2020. Then they build eight regression models using the training data that consisted of NIFTY 50 index records during December 29, 2014 till December 28, 2018. Using these regression models, we predicted the open values of NIFTY 50 for the period December 31, 2018 till July 31, 2020. The results clearly indicate that the LSTM-based univariate model that uses one-week prior data as input for predicting the next week's open value of the NIFTY 50 time series is the most accurate model.

Chapter 3: METHODOLOGY

3.1. System Block Diagram

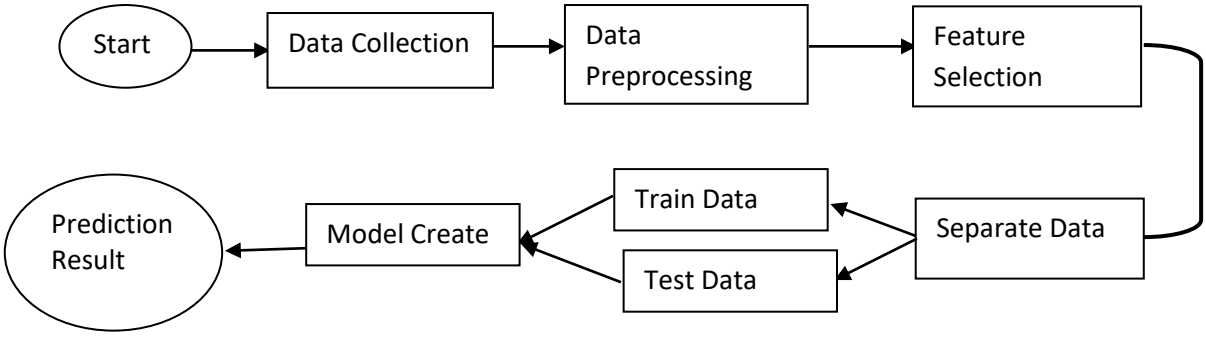

Figure 3.1 Block Diagram of Stock Prediction using LSTM model

3.1.1. Data Collection and Preprocessing

The study utilizes historical stock price data sourced from NepseAlpha, since the API of official website of Nepal Stock Exchange (NEPSE) is not publicly available. Data preprocessing involves several steps to prepare the data for analysis and modelling: Filtering: The dataset is filtered to include only the 'Close' price column, as the focus of the study is on predicting closing stock prices.

Scaling: The closing prices are scaled using the MinMaxScaler from the sklearn library to normalize the values between 0 and 1. This step is crucial for the efficient training of the LSTM model.

Training and Test Set Split: The dataset is divided into training and test sets, with 85% of the data used for training and the remaining 15% for testing. This split ensures that the model can be evaluated on unseen data.

Sequence Creation: The training data is transformed into sequences of 60 days each, which serve as the input features (X) for the LSTM model. The corresponding target (Y) for each sequence is the closing price on the 61st day.

3.1.2. LSTM Model Architecture

The study employs a Long Short-Term Memory (LSTM) neural network model for stock price prediction. The LSTM model is chosen for its ability to capture temporal dependencies and handle time-series data effectively [10]. Figure 3 shows the visualization of the architecture. The model architecture comprises the following layers. Input Layer: The input layer is configured to accept sequences of 60 days' worth of scaled closing prices. LSTM Layers: Two LSTM layers are used, with the first layer having 128 units and returning sequences to feed into the subsequent layer. The second LSTM layer has 64 units and does not return sequences. Dense Layers: Following the LSTM layers, two dense layers with 25 and 1 unit(s) respectively are added. The final dense layer outputs the predicted closing price. Compilation: The model is compiled using the Adam optimizer and mean squared error (MSE) as the loss function.

LSTM cell

The whole computation can be defined by a series of equations as follows:

it = σ (W i H + bi) (1)

ft = σ (W f H + bf) (2)

ot = σ (W oH + bo) (3)

ct = tanh (W c H + bc) (4)

mt = ft mt - 1 + it (5)

ht = tanh (ot mt) (6)

3.1.3. Model Training and Evaluation

The LSTM model is trained on the prepared training dataset with a batch size of 1 and 1 epoch. The training process involves adjusting the model's weights to minimize the loss function, thereby improving the accuracy of predictions . Model evaluation is conducted on the test dataset, where the

model's predictions are compared against the actual closing prices. The root mean squared error (RMSE) is calculated to quantify the model's prediction accuracy . Lower RMSE values indicate better model performance, where n is the number of observations, yi is the actual value of the i-th observation, and yi is the predicted value of the i-th observation

3.2. Description of working principle

LSTMs work in a three-step process.

- The first step in LSTM is to decide which information to be omitted from the cell in that particular time step. It is decided with the help of a sigmoid function. It looks at the previous state (ht-1) and the current input xt and computes the function.
- There are two functions in the second layer. The first is the sigmoid function, and the second is the tanh function. The sigmoid function decides which values to let through (0 or 1). The tanh function gives the weightage to the values passed, deciding their level of importance from -1 to 1.
- The third step is to decide what will be the final output. First, you need to run a sigmoid layer which determines what parts of the cell state make it to the output. Then, you must put the cell state through the tanh function to push the values between -1 and 1 and multiply it by the output of the sigmoid gate.

With this basic understanding of LSTM, you can dive into the hands-on demonstration part of this tutorial regarding stock price prediction using machine learning.

3.3. System Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) model that is most suitable for the LSTM-based stock market prediction project is the **Iterative and Incremental model**. Here's why this model is appropriate for your project:

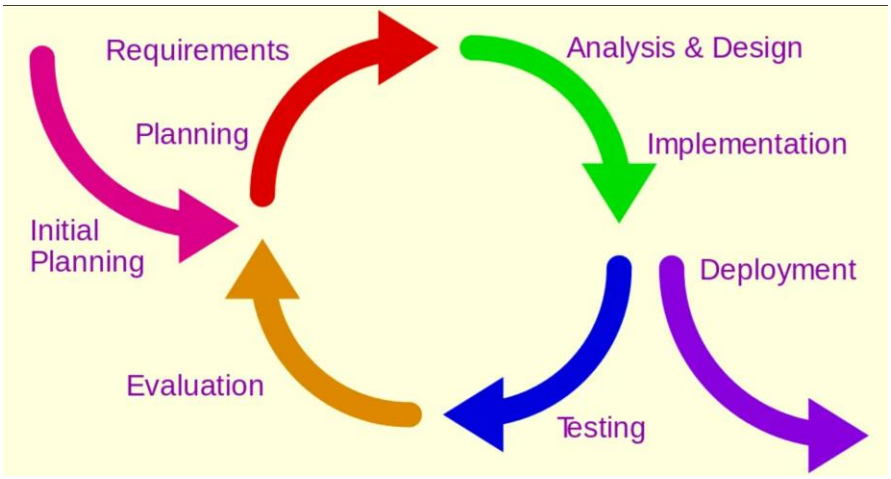

figure 3.3.1 Iterative and Incremental Model

Reasons of using Iterative and Incremental Model:

3.3.1. Phased Approach: The project can be divided into multiple phases, such as data collection, data preprocessing, model development, model evaluation, and documentation. Each phase can be iteratively refined based on feedback and results from previous phases.

3.3.2. Feedback Loops: Regular evaluations of the model's performance allow for adjustments and improvements. For instance, after developing an initial LSTM model, the team can analyze its accuracy and modify the architecture or hyperparameters as necessary, iterating until satisfactory performance is achieved.

3.3.3. Risk Management: By breaking the project into smaller, manageable increments, risks can be identified and addressed early. For example, if data quality issues arise during the preprocessing phase, they can be resolved before moving on to model training.

3.3.4. Flexibility: This model allows for flexibility in responding to new findings or changes in project scope. If new data sources or features become available that could enhance the model, these can be incorporated into the project without major disruptions.

3.4. Assumptions

- **Data Sufficiency for Training**: The amount of historical data needed for more accurate prediction will be much higher. So, we are assuming the data we collect must be of more than 5 years.
- **Closing Prices only**: Since we are working on stock prices and in context of Nepal, there is no intraday, so we are taking only closing prices of the stock as data.
- While writing the code and during training and testing of our application, we will be taking data of only one stock and use it as the test subject.

3.5. Hardware and Software Required

Hardware Requirements: (a) Computer with Intel 3 or higher processor with at least 4GB of RAM and 128GB of SSD.

Software Requirements:

- Google Colab: We will be using Google Colab for creating python project to work as a team
- Python: As a programming language, since it is best for machine learning projects

  Tools required to do this project:

  - **TensorFlow**: python library which is used to create Neural Networks
  - **Matplotlib**: python library to plot the chart from given stock closing prices
  - **NumPy**: python library used for numerical computing and interoperability of linear algebra
  - **Pandas**: python library for data analysis and manipulation

3.6. Machine Learning Algorithms

Long Short-Term Memory (LSTM) algorithm is used in this project for predicting stock prices. Long Short-Term Memory (LSTM) networks are often considered better for stock market prediction compared to traditional models and other machine learning techniques because of following reasons:

- Ability to capture large sequential data and can learn long-term dependencies.
- It can handle Non-Linear and complex Patterns in the chart or prices
- It is more suited for time series data
- It has ability to learn from noise and volatility, since stock prices are often noisy and volatile

3.7. Data Collection

Since the API of official website of Nepal Stock Exchange (NEPSE) is not publicly available, all the necessary historical data will be collected from a website called NepseAlpha, a private company work in the field of stock market. We will be collected the OHLC(Open, High, Low, Closing) prices of the day for that stock of minimum 5 years.

3.8. Implementation Plan

Step 1: Python will be used as programming language, and VS Code and Google Colab is used as IDEs.

Step 2: Data are collected through Nepsealpha using Pandas

Step 3: Data will be manipulated and Pre processing on them will be accomplish through NumPy

Step 4: Model will be created in LSTM algorithm, using Keras from TensorFlow

Step 5: Data Training and Testing

Step 6: Data Validation, Prediction, and Evaluation

Step 7: Optimization and UI/UX design

Step 8: Documentation

Chapter 4: Expected Output

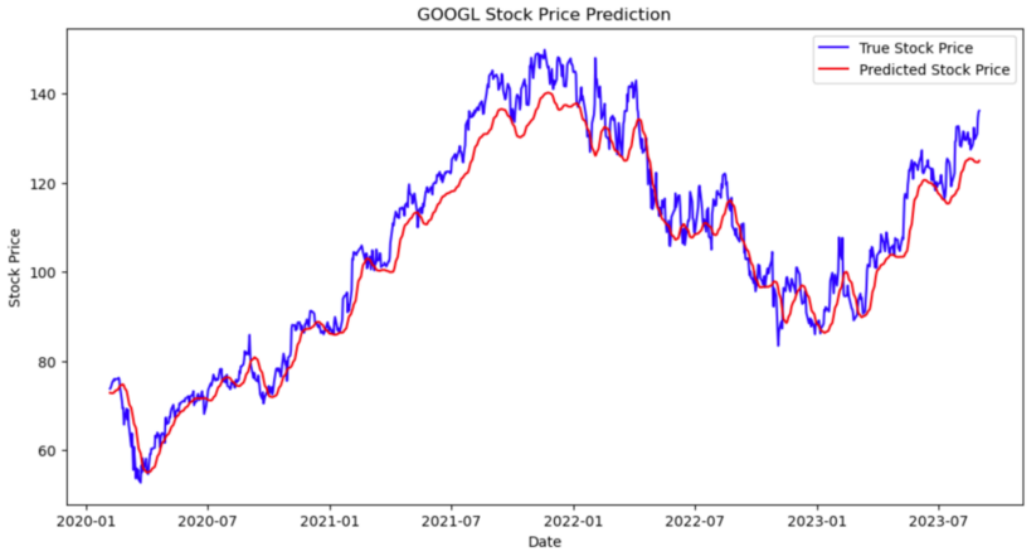

Figure 4.1. Actual Prive vs Predicted Price

After completion of our Model, we can draw the actual price and our predicted price alongside the historical data of that stock or index, of predict the price of upcoming 30 days.

4.1.1 Performance Metrics

Mean Squared Error (MSE):
MSE measures the average of the squared differences between the predicted and actual values. It penalizes larger errors more severely than smaller ones, making it useful for detecting major deviations in stock price predictions.

Formula:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \tilde{y}_i)^2$$

Goal: Minimize the MSE to improve the accuracy of predictions.

Interpretation: Lower MSE indicates better model performance.

Chapter 5: Project Schedule

5.1. Schedule

Table 5.1 Gantt Chart

S.N.	Tasks	Month			
1.	Discussion & Planning	Sept 20 - 24			
2.	Project Setup & Literature Review	Sept 26 -	Oct 10		
3.	Data Collection & Preprocessing			Nov 11 – 20	
4.	Model Design & Development			Nov 30 – Dec 20	
5.	Model Training & Evaluation			Dec 20 – Jan 10	
6.	Optimization			Jan 10 – Jan 20	
7.	Documentation, Report Writing & Submission				Jan 21 Afterwards

Chapter 6: Feasibility Analysis

6.1. Technical Feasibility

6.1.1. Data Availability

Stock Market Data, including historical prices, trading volumes and technical indicators are readily available from platforms like NepseAlpha, Nepal Stock Exchange (NEPSE) etc. This provides strong foundation for training and testing this model.

6.1.2. Machine Learning Frameworks

Libraries like TensorFlow, Keras and Pandas offers highly efficient tools for building LSTM model and train it to build a efficient model. We don't need very High end computer to do all these tasks.

6.1.3. Scalability and Optimization

LSTM models can be optimized using advanced techniques like grid search for HPT. The model can be also be scaled to handle more data and features, like sentiment analysis or macroeconomic indicators, making it further more efficient.

6.2. Operational Feasibility

6.2.1. Team Expertise

Each of the member in our team are directly or indirectly involved in the Stock Market for last 3 years. That makes us more knowledgeable in this field, which helps to build the model efficient.

6.2.2. Data Management

Handling stock market data, which is generally well-structured and publicly available, is operationally feasible. The team can source historical stock data and technical indicators from free APIs

6.3. Economic Feasibility

6.3.1. Infrastructure & Tools

The project does not require heavy infrastructural investments. Development can be carried out using open-source tools and cloud-based platforms such as Google Colab, which offers free GPU access for training

machine learning models. This eliminates the need for expensive local hardware and allows the team to focus on model development without worrying about infrastructure costs.

6.3.2. Cost of Development

The primary software tools and libraries needed for the project (e.g., Python, TensorFlow, Keras, pandas) are open-source and free to use. This eliminates any significant software licensing costs. Historical stock market data can be sourced from free APIs

6.3.3. Hardware and Infrastructure:

The project can leverage cloud computing platforms like Google Colab, which provides free access to GPUs for training deep learning models. This reduces the need for costly local hardware, making it economically feasible to train the models efficiently.

6.4. Time Feasibility

6.4.1. Time and Resource Management

The outlined project timeline (16-17 weeks) is realistic and can be managed within a typical academic year. By breaking down the project into clear stages—such as data collection, model development, evaluation, and optimization—the workload can be distributed efficiently among team members. This ensures that the project remains on track and aligns with academic schedules and deadlines.

6.4.2. Project Timeline

The project is planned over a typical academic semester, spanning approximately 16-17 weeks. This timeframe is generally sufficient for completing similar projects in engineering programs.

6.4.3. Realistic Expectations

Setting realistic goals for model performance and project scope is essential to avoid time overruns. The team should prioritize key deliverables and outcomes, focusing on achieving a functional model rather than striving for perfection in every aspect.

Chapter 7: CONCLUSION

In this study, we explored the application of Long Short-Term Memory (LSTM) networks for stock market analysis and prediction, focusing on four major stocks: Everest Bank Limited (EBL), Chilime Hydropower Company Limited (CHCL) and Standard Chartered Bank Limited (SCB). Our research demonstrated the potential of LSTM models in capturing complex patterns and trends in stock price movements, leading to reasonably accurate predictions of future prices. The findings of this study hold significant implications for both academia and the finance industry.

For investors and traders, the insights gained from our model could aid in making more informed decisions, potentially leading to improved investment strategies and portfolio management.

In conclusion, our study underscores the potential of LSTM networks in stock market analysis and prediction, offering valuable insights for both researchers and practitioners in the field of finance. As machine learning continues to evolve, its application in financial markets holds the promise of unlocking new opportunities for understanding and predicting stock price movements.

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